

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Report

Executive Summary

This report evaluates `deepseek/deepseek-v4-pro` across Infron and OpenRouter in prompt-caching workloads. The experiment uses only `/v1/chat/completions`, keeps model/platform default reasoning and thinking behavior, and keeps prompt-length stratification enabled.

After A/B filtering, the analysis retains **791 paired samples** and **3164 request-level observations**. It excludes **18 records**.

Core finding: Infron leads observed cost in every routing mode and leads E2E latency overall. OpenRouter leads throughput and Streaming TTFT in every routing mode. Cache-rate winners are close and vary by routing mode.

Quality Gate

Item	Value
Model	<code>deepseek/deepseek-v4-pro</code>
API protocol	<code>/v1/chat/completions</code> only
Groups / rounds	4 groups x 50 rounds
Prompt tiers	<code>short</code> about 1500, <code>medium</code> about 8000, <code>long</code> about 32000 tokens
Reasoning / thinking	Default platform/model behavior; no explicit disable parameter
Pairing rule	strict sort/group/round pair with first/second usage.prompt_tokens deltas <= 50
Retained pairs	791
Request-level rows	3164
Excluded records	18

Route–Level Result Matrix

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	200	Infron 90.52% / OpenRouter 95.98%	Infron \$1.205367 / OpenRouter \$2.110727	Infron 2.805 tok/s / OpenRouter 39.962 tok/s	Infron 4557.65 ms / OpenRouter 10533.90 ms	Infron 4139.22 ms / OpenRouter 3138.79 ms
Price First	198	Infron 99.44% / OpenRouter 98.26%	Infron \$0.337858 / OpenRouter \$1.693868	Infron 3.197 tok/s / OpenRouter 38.359 tok/s	Infron 4023.15 ms / OpenRouter 10514.30 ms	Infron 3350.61 ms / OpenRouter 3210.02 ms
Latency First	196	Infron 91.27% / OpenRouter 92.33%	Infron \$1.109211 / OpenRouter \$1.780845	Infron 2.949 tok/s / OpenRouter 39.703 tok/s	Infron 4305.92 ms / OpenRouter 10042.08 ms	Infron 3863.79 ms / OpenRouter 3191.40 ms
TTFT First	197	Infron 98.59% / OpenRouter 99.45%	Infron \$0.589968 / OpenRouter \$1.442047	Infron 2.790 tok/s / OpenRouter 36.153 tok/s	Infron 4588.39 ms / OpenRouter 10079.76 ms	Infron 4099.19 ms / OpenRouter 3196.65 ms

API Protocol Record

API protocol	Endpoint	Note
chat_completions	/v1/chat/completions	This run keeps only the standard Chat Completions protocol; it does not include /v1/messages or /v1/responses .

Prompt–Length Stratified Cache Performance

Tier	Pairs	Token cache hit rate	Call cache hit rate	E2E latency	TTFT	Reasoning tokens
short		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
medium		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
long		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0

Reasoning / Thinking Observation

This run does not explicitly disable or force reasoning/thinking. Reasoning tokens are treated as observed telemetry.

Routing mode	Infron reasoning tokens	OpenRouter reasoning tokens	Infron avg/request	OpenRouter avg/request
Throughput First	0	162216	0.0000	405.5400
Price First	0	153691	0.0000	388.1086
Latency First	0	150585	0.0000	384.1454
TTFT First	0	138211	0.0000	350.7893

Reproducibility References

Artifact	GitHub path	SHA256 / Notes
Summary	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/summary.json	0984289dd19dcfd12644cbe0d4e2473b57ee98cf65ced71209d32710e8
Paired dataset CSV	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_pairs.csv	858cdad534209908d7dbce59a92205040a3a008e639bf860d527b4358b
Request-level dataset JSONL	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_requests.jsonl	c1958ba57cc831fad6677f10937e6ce984137f351f2e7b1f1f05e34558
Excluded-record audit	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/records_excluded.json	104a6a0249e8026ea16e841ebdff3e855e6a526297db4a5ef6441f3999

Artifact	GitHub path	SHA256 / Notes
Benchmark runner source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py	a485c5116e99375e6b171d0328e5a423babd1ee6b206c72f54053a3a04
HTML report renderer source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/render_glm52_deepseek_style_report.py	e5bb0d32361d70ee688e0d0a0da3302a1e542e33e53876c905814eb5c2
Test source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/test_rerun_routing_sort_cache_cost_ab.py	c086ddc5d0a9a91eba82b7e8767d7bddf2f1ca4a28af0a87056b702882
Reports directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports	Bilingual HTML / Markdown / PDF reports
Data directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data	Pair data, request telemetry, summary, and excluded records
Manifest	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/metadata/manifest.json	File sizes and SHA-256 checksums

Dataset reference: `business_representative` built-in representative business templates; request-level export is `benchmark_requests.jsonl`.

Online HTML: Chinese https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_deepseek-v4-pro_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.zh.html; English https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_deepseek-v4-pro_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.en.html.